

What is vLLM?

vLLM is an **open-source LLM inference engine** built for **high-throughput, low-latency, and memory-efficient serving** of large language models. It's designed to run LLMs (like LLaMA, Mistral, Falcon, GPT-like models) **faster and cheaper** than naive PyTorch or Hugging Face serving.

Key Features

Feature	What it means	Why it matters
PagedAttention	Custom attention mechanism that stores KV cache in a paged format (like virtual memory)	Enables running long context (e.g., 32k+ tokens) efficiently without running out of GPU memory
Continuous Batching	Dynamically adds new requests into a running batch without waiting for old ones to finish	Drastically improves GPU utilization and throughput for APIs with many concurrent users
Tensor Parallelism	Splits model layers across multiple GPUs	Lets you serve huge models (70B+) on multiple GPUs
OpenAI-Compatible API	Built-in REST API that mimics OpenAI's API schema	Lets you switch from OpenAI API to self-hosted LLM with zero code changes in clients
Multi-LoRA Support	Load a base model and apply different LoRA adapters at runtime	Great for serving multiple fine-tuned variants without duplicating GPU memory
Streaming Support	Send partial outputs while the model is still generating	Faster perceived response times for users

Why vLLM is Popular

- Performance** — Higher throughput than Hugging Face `transformers` inference.
- Efficiency** — Can serve **more users per GPU**, lowering costs.
- Scalability** — Handles large models and long prompts well.
- Ease of Integration** — Works as a drop-in replacement for OpenAI API.
- Deployment Options** — Can run locally, on-prem, or in the cloud.

Example Use Case

Let's say you want to serve LLaMA-3-70B locally instead of paying OpenAI.
With vLLM:

```
vllm serve meta-llama/Meta-Llama-3-70B-Instruct \
  --port 8000 \
  --tensor-parallel-size 4 \
  --gpu-memory-utilization 0.9 \
  --max-model-len 8192
```

Now you can query it:

```
curl http://localhost:8000/v1/chat/completions \
  -H "Content-Type: application/json" \
  -d '{
    "model": "meta-llama/Meta-Llama-3-70B-Instruct",
    "messages": [{"role": "user", "content": "Tell me a joke"}],
    "max_tokens": 50
  }'
```